PROJECT REPORT

SMART RESUME ANALYZER

NAME : SUJAL SONI

REG. NO. : 25BAI10315

COURSE NAME : BTECH [AI / ML]

FACULTY NAME: DR. GARIMA JAIN

DATE OF SUBMISSION: 29/03/2026

1. Executive Summary

Skill Lens is an intelligent Resume Analyzer application designed to help users evaluate and improve their resumes using Artificial Intelligence and Machine Learning techniques. The system analyzes resumes, extracts key information, and provides insights such as skill gaps, keyword matching, and job compatibility.

The application aims to simplify the hiring preparation process by giving users actionable feedback to enhance their resumes and increase their chances of selection.

2. Problem Statement

Many job seekers face challenges such as:

- Creating ATS-friendly resumes
- Identifying missing or weak skills
- Matching resumes with job descriptions
- Understanding industry requirements
- Getting real-time feedback on resume quality

3. Functional Requirement

a. User Account Management

- Users can register and login securely
- Users can upload and manage resumes
- Password reset functionality

b. Resume Upload & Parsing

- Users can upload resumes (PDF/DOCX)
- System extracts text using NLP
- Identify sections like skills, education, experience

c. Skill Analysis

- Extract technical and soft skills
- Compare with job requirements
- Suggest missing skills

d. Job Matching

- Match resume with job description
- Provide compatibility score
- Highlight matched & missing keywords

e. Dashboard / Summary

- Resume score display
- Skill gap analysis
- Suggestions for improvement
- Graphical insights (charts)

4. NON FUNCTIONAL REQUIREMENTS

Performance

- Response time < 2 seconds
- Fast resume parsing

Usability

- Simple UI for upload and results
- Easy navigation

Reliability

- Accurate data extraction
- Error handling

Security

- Secure resume storage
- Data privacy protection

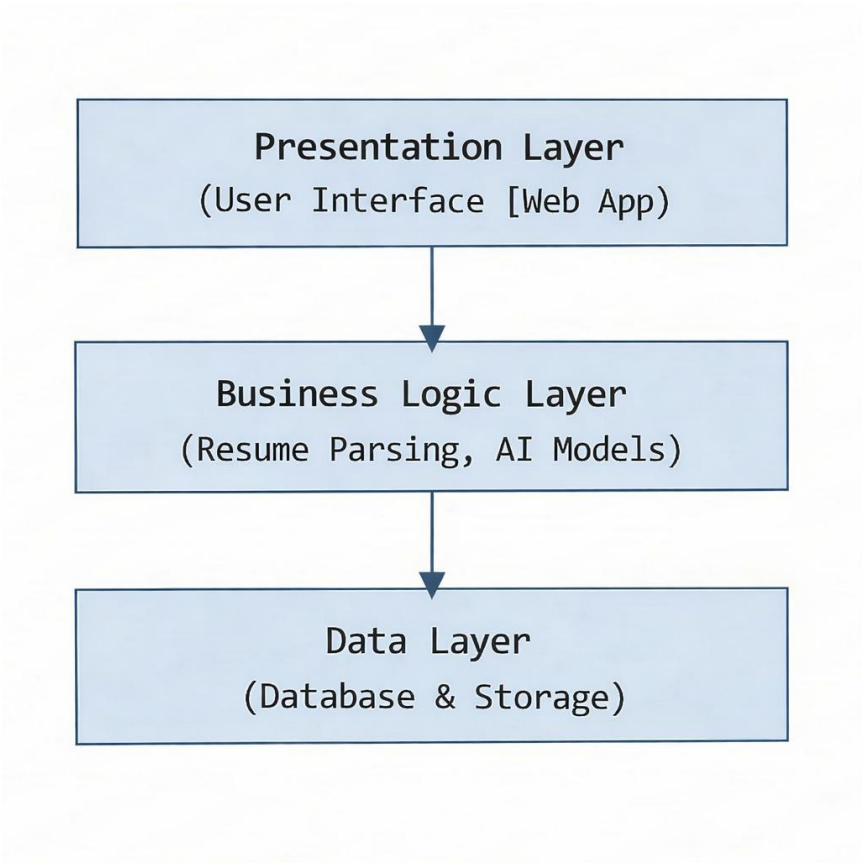
Scalability

- Support multiple users
- Handle large resume database

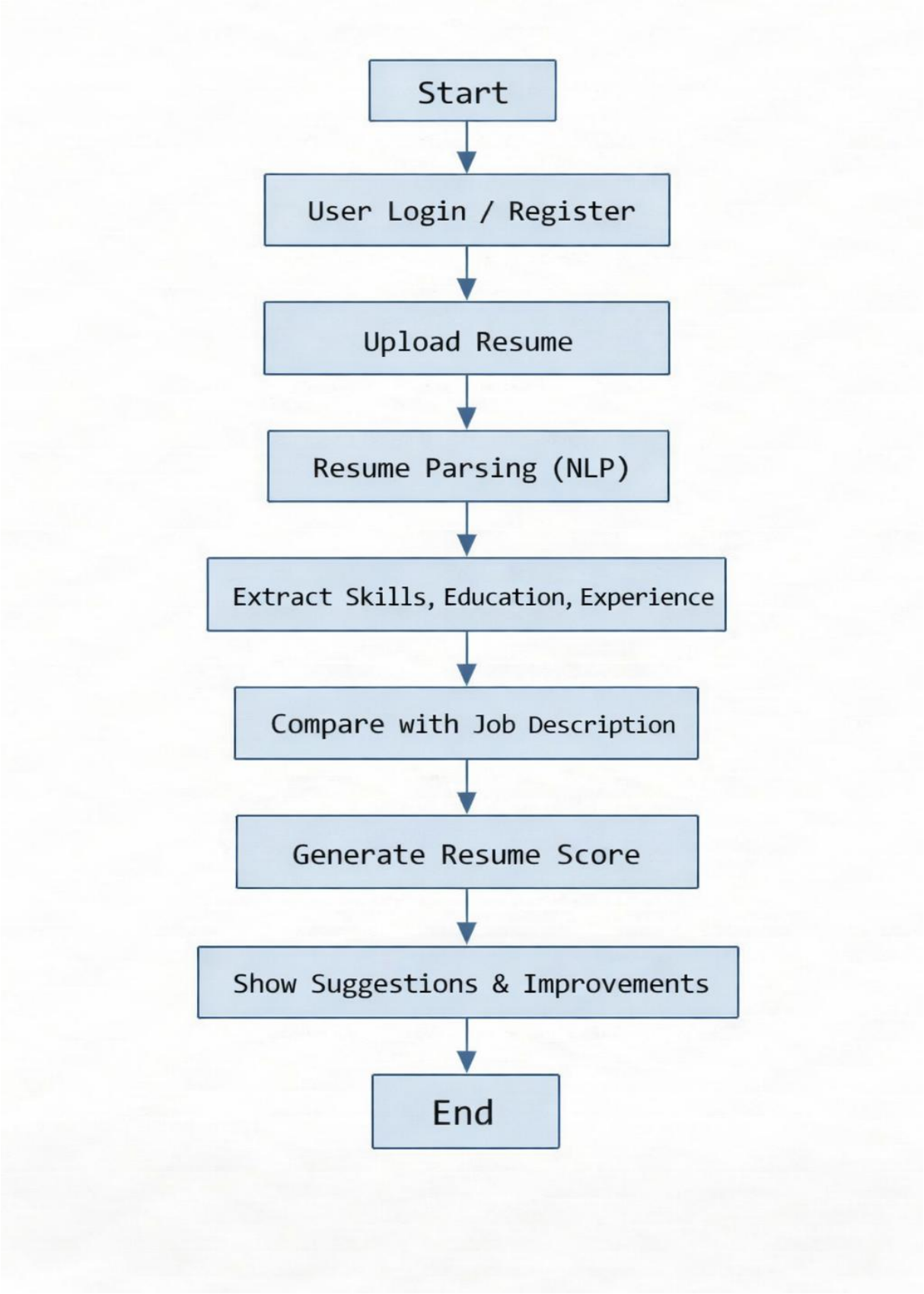
5. System Architecture

The system follows a **three-tier architecture**:

1. Presentation Layer → User Interface (Web App)
2. Business Logic Layer → Resume Parsing, AI Models
3. Data Layer → Database & Storage



6. Design Diagram



7. Features and Functionality

Core Features

1. User Management

- Registration & login
- Profile management

2. Resume Analysis

- Resume parsing using NLP
- Section detection
- Keyword extraction

3. Skill Matching

- Skill extraction
- Match with job roles
- Suggest improvements

4. Analytics and Reports

- Resume score
- Skill gap charts
- Keyword match percentage

5. Dashboard

- Resume insights
- Suggestions
- Score visualization

Additional Features

- ATS Score Checker
- Resume Formatting Suggestions
- Job Recommendation System
- Export Report (PDF)
- Dark Mode

Key Screens

1. Login Page
2. Resume Upload Page
3. Dashboard
4. Analysis Report Page
5. Suggestions Page

8. Design Principles

- Simple & Clean UI
- User-friendly
- Responsive design
- AI-driven insights

9. Testing Strategy

- Unit Testing
- Integration Testing
- End-to-End Testing
- Performance Testing

10. Technical Improvements

- AI-based Resume Scoring
- Advanced NLP Models
- Cloud Integration
- Real-time Job APIs

11. Conclusion

Skill Lens provides an efficient and intelligent solution for resume analysis. It helps users improve their resumes, identify skill gaps, and increase their chances of getting selected in job applications.